

Network

A network is a system that allows interconnection of multiple devices (nodes) such as computers, servers, printers, and other hardware, enabling them to communicate and share resources, like data, applications, and internet connections.

Types of Networks

Type	Description	Range
PAN (Personal Area Network)	Very small network, usually around a person.	Up to 10 meters
LAN (Local Area Network)	Network in a small area such as a building or office.	Up to a few kilometers
MAN (Metropolitan Area Network)	Covers a city or a campus.	Up to 50 km
WAN (Wide Area Network)	Covers large geographical areas; example: the Internet.	Unlimited

Key Components of a Network

- **Nodes:** Devices connected in the network (e.g., computers, routers, printers).
- **Links:** Physical or logical connections between nodes (e.g., cables, wireless).
- **Switches:** Connect devices within a LAN and manage data traffic.
- **Routers:** Connect different networks, such as LAN to WAN.
- **Protocols:** Rules governing data communication (e.g., TCP/IP, HTTP).

Network Goals

1. Resource Sharing – e.g., sharing files, printers.
2. Communication – through emails, chats, VoIP, etc.
3. Centralized Management – through servers.
4. Data Security & Backup – stored centrally and securely.

Network Topologies

A network topology is the physical or logical layout of how nodes and connections (links) are arranged in a network.

There are two types:

- **Physical topology:** Actual layout of cables and devices.
- **Logical topology:** How data flows within the network.

Types of Network Topologies

1. Bus Topology

Structure: All devices are connected to a single central cable (backbone).

- **Advantages:**
 - Easy to implement and extend.
 - Requires less cable than other topologies.
- **Disadvantages:**
 - A break in the backbone disables the entire network.
 - Performance degrades as more devices are added.

2. Star Topology

Structure: All nodes are connected to a central device (switch/hub).

- **Advantages:**
 - Easy to manage and troubleshoot.
 - Failure in one cable doesn't affect others.
- **Disadvantages:**
 - Failure of central hub disables entire network.
 - More cable required.

3. Ring Topology

Structure: Each device connects to two others, forming a ring.

- **Advantages:**
 - Data flows in one direction (or two in dual ring).
 - Predictable performance with fewer collisions.
- **Disadvantages:**
 - One failure affects entire ring (unless dual ring).
 - Difficult to add/remove devices.

4. Mesh Topology

Structure: Each node is connected to every other node.

- **Advantages:**
 - Highly reliable (redundant paths).
 - Failure of one link doesn't affect the network.
- **Disadvantages:**
 - Very expensive (many cables and ports).
 - Complex setup and maintenance.

5. Tree Topology

Structure: Hierarchical structure combining multiple star topologies.

- **Advantages:**
 - Scalable and easy to manage.
 - Fault isolation is easy.
- **Disadvantages:**
 - Dependent on backbone (root node).
 - Complex cabling.

6. Hybrid Topology

Structure: Combination of two or more topologies (e.g., star-bus).

- **Advantages:**
 - Flexible and scalable.
 - Customized to organizational needs.
- **Disadvantages:**
 - Complex to design and maintain.
 - Expensive.

Topology	Cost	Reliability	Scalability	Cabling
Bus	Low	Low	Low	Minimal
Star	Medium	Medium	High	High
Ring	Medium	Low	Medium	Medium
Mesh	High	Very High	High	Very High
Tree	Medium	Medium	High	High
Hybrid	High	High	Very High	High

Network Classifications

Networks can be classified in several ways depending on **size**, **function**, **access control**, and **data transmission techniques**.

Classification Based on Network Architecture:

Architecture	Description	Examples
Peer-to-Peer (P2P)	All devices (peers) have equal status and share resources directly.	File sharing apps, home networks
Client-Server	Centralized server provides resources/services to clients.	Web servers, database servers
Hybrid	Combination of both client-server and peer-to-peer models.	Corporate environments

Classification Based on Transmission Technology:

Type	Description	Example
Broadcast Networks	Single communication channel is shared by all devices; messages are sent to all nodes.	Traditional Ethernet
Point-to-Point Networks	Dedicated communication link between two devices.	Leased line, microwave links

Classification Based on Ownership/Access:

Type	Description	Examples
Private Network	Owned and operated by a single organization.	Enterprise LAN
Public Network	Open to public use.	Internet
Virtual Private Network (VPN)	Secure private connection over a public network.	Remote corporate access

Classification Based on Communication Direction:

Type	Description
Simplex	Data flows in one direction only. (e.g., TV broadcast)
Half-Duplex	Data flows in both directions, but only one at a time. (e.g., walkie-talkie)
Full-Duplex	Data flows in both directions simultaneously. (e.g., phone calls)

Network Protocol

A network protocol is a set of rules and conventions that govern how data is transmitted, formatted, and processed across a network. It ensures reliable, efficient, and secure communication between devices (nodes).

Why Are Protocols Important?

- Ensure **interoperability** between different devices and systems.
- Define **data formats**, **error handling**, and **message acknowledgment**.
- Enable **standardized communication** over the internet and other networks.
- Handle **data transmission**, **routing**, **security**, **congestion control**, and more.

Types of Network Protocols

Network protocols are generally categorized based on their **function** in the network. Below are the most common types:

1. Communication Protocols

Protocol	Description
TCP (Transmission Control Protocol)	Ensures reliable, connection-oriented data transmission. Handles error checking and retransmission.
UDP (User Datagram Protocol)	Connectionless and faster, but less reliable than TCP. Used in real-time apps like video streaming.
IP (Internet Protocol)	Handles addressing and routing of data packets across networks. IPv4 and IPv6 are two versions.
ICMP (Internet Control Message Protocol)	Used for diagnostic and error messages (e.g., ping).
ARP (Address Resolution Protocol)	Resolves IP addresses to MAC (hardware) addresses within a LAN.

2. File Transfer Protocols

Protocol	Description
FTP (File Transfer Protocol)	Used for transferring files between computers over a TCP network.
SFTP (Secure FTP)	FTP over SSH, providing encrypted file transfer.
TFTP (Trivial FTP)	A simpler, lightweight version of FTP without authentication.

3. Email Protocols

Protocol	Description
SMTP (Simple Mail Transfer Protocol)	Sends email from client to server or between servers.
POP3 (Post Office Protocol v3)	Downloads email from server to client, then deletes it from the server.
IMAP (Internet Message Access Protocol)	Keeps email on the server and allows multiple device access.

4. Web and Internet Protocols

Protocol	Description
HTTP (HyperText Transfer Protocol)	Foundation of web communication (non-secure).
HTTPS (HTTP Secure)	HTTP over SSL/TLS for secure communication.
DNS (Domain Name System)	Resolves domain names (like www.google.com) into IP addresses.

5. Remote Access Protocols

Protocol	Description
SSH (Secure Shell)	Securely connects to a remote computer over an unsecured network.
Telnet	Used for remote login, but not secure (plaintext).
RDP (Remote Desktop Protocol)	Microsoft's protocol for remote desktop access.

6. Network Management Protocols

Protocol	Description
SNMP (Simple Network Management Protocol)	Monitors and manages network devices like routers and switches.
NTP (Network Time Protocol)	Synchronizes clocks across networked devices.

OSI Model and Protocol Mapping

OSI Layer	Common Protocols
Layer 7 – Application	HTTP, FTP, SMTP, DNS, SNMP
Layer 6 – Presentation	SSL, TLS, MIME
Layer 5 – Session	NetBIOS, PPTP
Layer 4 – Transport	TCP, UDP
Layer 3 – Network	IP, ICMP, IGMP
Layer 2 – Data Link	Ethernet, ARP, PPP
Layer 1 – Physical	Ethernet (physical layer), DSL, Fiber, Wi-Fi signals

Layered Network Architecture

Layered network architecture is a design approach in which the network functions are divided into a series of layers, each responsible for specific tasks. Each layer provides services to the layer above it and receives services from the layer below it.

Purpose

- To simplify network design by breaking complex processes into manageable components.
- To allow modularity, meaning changes in one layer don't require changes in others.
- To provide standardization so that hardware and software from different vendors can work together.
- To enable interoperability between different systems and technologies.

Key Principles

1. **Layer Independence** – Each layer works independently, with well-defined interfaces.
2. **Encapsulation** – Higher layers don't need to know *how* lower layers implement their functions.
3. **Standard Interfaces** – Communication between layers is done via standard protocols.
4. **Service and Protocol Separation** –
 - **Service:** What a layer does for the layer above it.
 - **Protocol:** Rules that govern communication between the same layer on different devices.

Common Layered Models

A. OSI (Open Systems Interconnection) Model

- **7 layers:**

- 1. Physical Layer – Deals with raw bit transmission over a medium (cables, signals).**
- 2. Data Link Layer – Handles framing, MAC addresses, error detection.**
- 3. Network Layer – Responsible for logical addressing (IP), routing.**
- 4. Transport Layer – Ensures reliable delivery, error correction (TCP, UDP).**
- 5. Session Layer – Manages sessions between applications.**
- 6. Presentation Layer – Data format translation, encryption, compression.**
- 7. Application Layer – Interfaces for user applications (HTTP, FTP, DNS).**

B. TCP/IP Model

- **4 layers (practical implementation of networking):**

- 1. Link Layer – Similar to OSI's Physical + Data Link layers.**
- 2. Internet Layer – Similar to OSI's Network layer (IP).**
- 3. Transport Layer – TCP, UDP.**
- 4. Application Layer – Combines OSI's session, presentation, application.**

Advantages

- **Modularity** – Easier troubleshooting and upgrades.
- **Interoperability** – Different vendor systems can work together.
- **Scalability** – Layers can be improved independently.

- **Standardization** – Reduces compatibility issues.
- **Flexibility** – Multiple protocols can operate at the same layer.

Disadvantages

- **Overhead** – Extra processing due to encapsulation at each layer.
- **Redundancy** – Some functions may be duplicated across layers.
- **Not Always Optimal** – Layer separation can reduce performance in high-speed networks.

Layer (OSI)	Function	Example Protocols
Application	User services & network apps	HTTP, FTP, SMTP, DNS
Presentation	Data translation, encryption	SSL, JPEG, ASCII
Session	Session establishment & termination	NetBIOS, RPC
Transport	Reliable delivery, error correction	TCP, UDP
Network	Logical addressing, routing	IP, ICMP
Data Link	Framing, physical addressing	Ethernet, PPP, Wi-Fi
Physical	Transmission of raw bits	Cables, radio waves

Overview of OSI Reference Model

The OSI (Open Systems Interconnection) Reference Model is a conceptual framework that standardizes the functions of a network into seven layers. It was developed by ISO (International Organization for Standardization) to promote interoperability between different networking systems.

Purpose

- To provide a universal standard for network communication.
- To help design, troubleshoot, and understand complex networks by dividing functions into layers.
- To allow different hardware, software, and vendors to work together seamlessly.

7 Layers of OSI

Layer No.	Layer Name	Main Function	Examples
7	Application	Interfaces with user applications	HTTP, FTP, SMTP
6	Presentation	Data translation, encryption, compression	JPEG, ASCII, SSL
5	Session	Establishes, manages, and terminates sessions	NetBIOS, RPC
4	Transport	Reliable delivery, error correction	TCP, UDP
3	Network	Logical addressing, routing	IP, ICMP
2	Data Link	Framing, physical addressing (MAC)	Ethernet, Wi-Fi
1	Physical	Transmission of raw bits over medium	Cables, signals

Key Features

- **Layered Approach** – Each layer performs a specific role and interacts with adjacent layers.
- **Encapsulation** – Data is wrapped with protocol information at each layer.
- **Protocol Independence** – Any protocol can work at a given layer if it meets the standards.
- **Logical Structure** – Separates *what* is done from *how* it is done.

Advantages

- **Standardization** promotes interoperability.
- **Easier troubleshooting** — faults can be isolated to a specific layer.
- **Modular design** — allows independent development of each layer.

Limitations

- **Theoretical model** — not used exactly in real networks.
- **Can introduce overhead** due to strict layering.

TCP/IP Protocol Suite

Full form: Transmission Control Protocol / Internet Protocol

The TCP/IP protocol suite is a set of communication protocols used for the Internet and similar networks.

It is based on a four-layer conceptual model that specifies how data should be packaged, addressed, transmitted, routed, and received.

Layers of TCP/IP Model

Layer No.	Layer Name	Main Functions	Example Protocols
4	Application Layer	Network services for user applications (combines OSI's application, presentation, and session layers)	HTTP, FTP, SMTP, DNS, Telnet
3	Transport Layer	End-to-end communication, error detection, flow control	TCP, UDP
2	Internet Layer	Logical addressing, routing of packets	IP (IPv4/IPv6), ICMP, ARP, RARP
1	Link Layer (or Network Access Layer)	Physical addressing, framing, and medium access control	Ethernet, Wi-Fi, PPP

Working Principle

- **Encapsulation:** Data from an application is passed down through layers, each adding a header (and sometimes a trailer).
- **Decapsulation:** At the receiving end, each layer processes and removes its header, passing the remaining data up.

Key Protocols

- **TCP (Transmission Control Protocol)** – Reliable, connection-oriented communication.
- **UDP (User Datagram Protocol)** – Faster, connectionless communication.
- **IP (Internet Protocol)** – Logical addressing & routing.

- **ICMP (Internet Control Message Protocol) – Error reporting & diagnostics (ping).**
- **ARP (Address Resolution Protocol) – Maps IP to MAC addresses.**
- **DNS (Domain Name System) – Translates domain names to IP addresses.**

Features

- **Interoperability – Works across heterogeneous networks.**
- **Scalability – Can connect millions of devices.**
- **Flexibility – Supports multiple applications and data types.**
- **End-to-End Communication – Ensures data is delivered to the right application on the right device.**

Advantages

- **Global standard for internetworking.**
- **Robust and fault-tolerant.**
- **Independent of underlying hardware.**

Disadvantages

- **Protocol overhead may reduce efficiency.**
- **No strict separation of services (compared to OSI model).**

Introduction to Data Communication

Data Communication refers to the exchange of data between two or more devices over a communication medium (like cables, radio waves, or fiber optics), using a set of protocols to ensure correct transmission.

Fundamental Components

Every data communication system has five basic components:

Component	Description
Message	The data or information to be communicated (text, numbers, images, video, etc.).
Sender	Device that sends the data (e.g., computer, phone, sensor).
Receiver	Device that receives the data.
Medium	Physical path between sender and receiver (copper wire, fiber optic, radio waves).
Protocol	Rules that govern communication (format, timing, error checking).

Characteristics of Effective Communication

For data communication to be effective, it must have:

- 1. Delivery – Data must reach the correct destination.**
- 2. Accuracy – Data must be delivered without alteration.**
- 3. Timeliness – Data must arrive when needed (especially for real-time applications).**
- 4. Jitter – Variation in packet arrival time should be minimal (important in audio/video).**

Data Communication Modes

4.1 Direction of Data Flow

- 1. Simplex – One-way communication only (e.g., keyboard → computer).**
- 2. Half-Duplex – Two-way communication, but one direction at a time (e.g., walkie-talkies).**
- 3. Full-Duplex – Two-way communication simultaneously (e.g., telephone).**

Data Communication Techniques

5.1 Analog and Digital Signals

- Analog Signal – Continuous waveform (e.g., human voice).**
- Digital Signal – Discrete levels (binary 0 and 1).**

Conversion techniques:

- Analog-to-Digital: Sampling, Quantization, Encoding.**
- Digital-to-Analog: Modulation.**

Transmission Modes

- 1. Serial Transmission – Bits sent one after another over a single channel.**
- 2. Parallel Transmission – Multiple bits sent simultaneously over multiple channels.**

Transmission Media

A. Guided Media (wired)

- Twisted Pair Cable**
- Coaxial Cable**
- Optical Fiber**

B. Unguided Media (wireless)

- **Radio Waves**
- **Microwaves**
- **Infrared**

Data Transmission Methods

- 1. Synchronous Transmission – Data sent in a continuous stream, synchronized by clock pulses.**
- 2. Asynchronous Transmission – Data sent one byte at a time with start/stop bits.**
- 3. Isochronous Transmission – Data delivered at a constant rate (for multimedia).**

Aspect	Options/Types	Examples
Flow Direction	Simplex, Half-Duplex, Full-Duplex	Keyboard, Walkie-talkie, Telephone
Signals	Analog, Digital	Voice, Binary data
Transmission	Serial, Parallel	USB, Printer cable
Media	Guided, Unguided	Fiber, Wi-Fi
Error Handling	Detection, Correction	CRC, ARQ

Switching in Networks

Switching is the process of forwarding data from the sender to the receiver across a network using intermediate nodes (switches or routers).

Switching determines **how data is routed** through the network.

Types of Switching Techniques

A. Circuit Switching

- **Concept:** A dedicated communication path is established between sender and receiver for the entire duration of the communication.
- **Steps:**
 1. **Establish** the circuit.
 2. **Transfer** the data.
 3. **Release** the circuit.
- **Characteristics:**
 - Fixed bandwidth for the session.
 - Low delay after connection is established.
 - Wastes resources if no data is being transmitted.
- **Example:** Traditional telephone networks.

Packet Switching

- **Concept:** Data is broken into packets, each packet is routed independently through the network.
- **Types:**
 1. **Datagram Packet Switching** – Each packet may take a different route; no fixed path.
 2. **Virtual Circuit Packet Switching** – A logical path is established before transmission, but data is still sent in packets.

- **Characteristics:**
 - More efficient use of network resources.
 - Packets can arrive out of order (reassembly needed).
 - Suitable for bursty data traffic.
- **Example:** The Internet.

Message Switching

- **Concept:** The entire message is sent from one node to the next (store-and-forward) without breaking it into smaller packets.
- **Characteristics:**
 - No dedicated path.
 - High delay for large messages.
 - Not suitable for real-time communication.
- **Example:** Telegraph systems.

Feature	Circuit Switching	Packet Switching	Message Switching
Path	Dedicated	Dynamic	Dynamic
Resource Use	Fixed allocation	Shared	Shared
Delay	Low (after setup)	Variable	High
Real-Time Support	Good	Fair (depends)	Poor
Example	PSTN	Internet	Telegraph

Network Access Mechanisms

Access mechanisms define **how multiple devices share a common communication medium** without interference.

Random Access

- Devices transmit when they have data, with collision detection or avoidance.
- **Examples:**
 1. **ALOHA** – Simple, send whenever ready; collisions cause retransmissions.
 2. **Slotted ALOHA** – Time is divided into slots; reduces collisions.
 3. **CSMA (Carrier Sense Multiple Access)** – Device listens to channel before sending.
 - **CSMA/CD** – Collision Detection (Ethernet).
 - **CSMA/CA** – Collision Avoidance (Wi-Fi).

Controlled Access

- Devices take turns using the medium.
- **Examples:**
 1. **Polling** – Master device asks each node in turn.
 2. **Token Passing** – A special frame (token) is passed; only the holder can transmit.
 - Example: Token Ring networks.

Channelization (Channel Partitioning)

- Medium is divided into separate channels for each user.
- **Techniques:**
 1. **FDMA (Frequency Division Multiple Access)** – Each user gets a unique frequency band.
 2. **TDMA (Time Division Multiple Access)** – Each user gets a time slot.

3. CDMA (Code Division Multiple Access) – Each user gets a unique code; signals share time and frequency.

Method	How It Works	Efficiency	Examples
Random Access	Transmit anytime, handle collisions	Good for light load	Ethernet, Wi-Fi
Controlled Access	Turn-taking	Good for heavy load	Token Ring, Polling
Channelization	Divide medium into channels	Constant	FDMA in radio, TDMA in GSM, CDMA in 3G

Real-Life Analogies

- **Circuit Switching** → Reserving a private lane on a highway for one trip.
- **Packet Switching** → Sending letters through postal service (each may take different routes).
- **Message Switching** → Shipping the entire package through one warehouse at a time.
- **Random Access** → People talking in a meeting without raising hands (possible interruptions).
- **Controlled Access** → People taking turns to speak.
- **Channelization** → Dividing a road into separate lanes.

Data Link Layer

The Data Link Layer (DLL) is the second layer in the OSI reference model, lying above the Physical Layer and below the Network Layer.

Its primary role is to ensure reliable transfer of data between two directly connected nodes over a physical medium.

- **Purpose:**

The Physical Layer transmits raw bits, but it cannot guarantee error-free delivery. The Data Link Layer adds necessary mechanisms to detect and correct errors, regulate data flow, and control access to the medium.

Major Functions of Data Link Layer

Framing: The process of dividing the data stream into manageable units called frames.

- **Why needed:** The Physical Layer deals with continuous streams of bits; frames provide boundaries so the receiver knows where a message starts and ends.
- **Framing Methods:**
 - **Character Count** – First field specifies frame length.
 - **Character Stuffing** – Special start/stop characters, with escape characters to avoid confusion.
 - **Bit Stuffing** – Special bit patterns mark frame boundaries; extra bits are inserted to prevent false detection.
 - **Physical Layer Coding Violations** – Use unused signal patterns to indicate frame boundaries.

Physical Addressing

- **The Data Link Layer uses MAC (Media Access Control) addresses to identify the source and destination devices within the same network.**
- **Example: Ethernet uses 48-bit MAC addresses.**

Error Detection and Correction

- **Error Detection:** Identifies whether data has been corrupted during transmission.
- **Techniques:**
 - **Parity Check (Even/Odd)**
 - **Cyclic Redundancy Check (CRC)**
 - **Checksum**
- **Error Correction:** Allows the receiver to recover the original data.
- **Techniques:**
 - **Automatic Repeat Request (ARQ)** – Retransmits data on error.
 - **Forward Error Correction (FEC)** – Corrects errors without retransmission.

Flow Control

- **Prevents the sender from overwhelming the receiver with too much data at once.**
- **Methods:**
 - **Stop-and-Wait** – Sender sends one frame, waits for acknowledgment before sending the next.
 - **Sliding Window** – Multiple frames can be sent before needing an acknowledgment; window size controls flow.

Access Control

- **In shared channels, the Data Link Layer decides which device gets to use the channel at a given time.**
- **Controlled by Media Access Control (MAC) sublayer.**
- **Techniques:**
 - **Random Access (CSMA/CD for Ethernet, CSMA/CA for Wi-Fi)**
 - **Controlled Access (Polling, Token Passing)**

Sub-layers of Data Link Layer

1. **Logical Link Control (LLC)** – Manages flow control, error detection, and provides interface to Network Layer.
2. **Media Access Control (MAC)** – Manages access to the physical transmission medium and handles MAC addressing.

Common Data Link Layer Protocols

Protocol	Description	Usage
Ethernet (IEEE 802.3)	Widely used LAN protocol, uses CSMA/CD for access control.	Wired LANs
Wi-Fi (IEEE 802.11)	Wireless LAN protocol, uses CSMA/CA.	WLAN
PPP (Point-to-Point Protocol)	Provides framing, authentication, error detection over serial links.	WAN links
HDLC (High-Level Data Link Control)	Bit-oriented protocol for point-to-point and multipoint links.	WAN
Frame Relay	High-speed packet-switching protocol.	WAN
ATM (Asynchronous Transfer Mode)	Cell-based switching for voice, video, data.	WAN

1. Point-to-Point Protocol (PPP)

- **Type:** Byte-oriented
- **Purpose:** Used to directly connect two nodes over serial links (e.g., DSL, ISPs).
- **Features:**
 - **Framing:** Uses a special flag byte 01111110 at start and end of frame.
 - **Error Detection:** 16-bit or 32-bit CRC.

- **Authentication:** Supports PAP (Password Authentication Protocol) and CHAP (Challenge Handshake Authentication Protocol).
- **Protocol Multiplexing:** Supports multiple Layer 3 protocols (IPv4, IPv6, IPX).
- **Advantages:**
 - Simple design.
 - Supports multiple network layer protocols simultaneously.
- **Example Usage:** Dial-up Internet, DSL connections.

2. High-Level Data Link Control (HDLC)

- **Type:** Bit-oriented
- **Standardized by:** ISO
- **Purpose:** General-purpose protocol for both point-to-point and multipoint links.
- **Features:**
 - **Framing:** Uses 01111110 as a flag pattern (bit stuffing used to avoid confusion).
 - **Modes:**
 1. **Normal Response Mode (NRM)** – Master-slave communication.
 2. **Asynchronous Balanced Mode (ABM)** – Peer-to-peer communication.
 3. **Asynchronous Response Mode (ARM)** – Rarely used.
 - **Error Control:** CRC-based error detection + ARQ for retransmission.

- **Advantages:**
 - **Reliable and supports both control and data frames.**
- **Example Usage: WAN connections, ISDN.**

3. Ethernet (IEEE 802.3)

- **Type: Frame-based LAN protocol.**
- **Access Method: CSMA/CD (Carrier Sense Multiple Access with Collision Detection).**
- **Features:**
 - **Frame Format: Contains preamble, destination & source MAC addresses, length/type, payload, and CRC.**
 - **Error Detection: CRC-32.**
 - **MAC Addressing: 48-bit unique identifier.**
- **Advantages:**
 - **Simple, low-cost.**
 - **Standardized and widely available.**
- **Example Usage: Office LANs, campus networks.**

4. Wi-Fi (IEEE 802.11)

- **Type: Wireless LAN protocol.**
- **Access Method: CSMA/CA (Collision Avoidance).**
- **Features:**
 - **Uses acknowledgment frames to confirm reception.**
 - **Includes security mechanisms like WPA/WPA2.**
 - **Frame structure includes management, control, and data frames.**

- **Error Handling:** Uses CRC for detection, retransmission for correction.
- **Advantages:**
 - Supports mobility.
 - High data rates (up to multi-Gbps in Wi-Fi 6).
- **Example Usage:** Home/office wireless networks, public hotspots.

5. Token Ring (IEEE 802.5)

- **Type:** LAN protocol using token-passing method.
- **Access Method:** Controlled access – a special frame called a token circulates in the network; only the station holding the token can transmit.
- **Advantages:**
 - No collisions.
 - Predictable access time.
- **Disadvantages:**
 - More complex and costly than Ethernet.
- **Example Usage:** Legacy IBM networks.

6. Frame Relay

- **Type:** WAN packet-switching protocol.
- **Purpose:** Designed for high-speed data transmission over reliable links.
- **Features:**
 - Minimal error correction (depends on underlying reliability).
 - Uses DLCI (Data Link Connection Identifier) to identify virtual circuits.

- **Advantages:**
 - **High throughput, low latency.**
- **Example Usage: Legacy WAN backbone connections.**

7. Asynchronous Transfer Mode (ATM)

- **Type: Cell-switching protocol.**
- **Cell Size: Fixed length of 53 bytes (5-byte header + 48-byte payload).**
- **Purpose: Supports voice, video, and data over the same network.**
- **Advantages:**
 - **Low latency.**
 - **Predictable performance.**
- **Example Usage: Telecom backbones, IPTV distribution.**

8. Logical Link Control (LLC – IEEE 802.2)

- **Part of: Upper sublayer of Data Link Layer.**
- **Purpose: Provides interface between Network Layer and MAC sublayer.**
- **Functions:**
 - **Multiplexing/demultiplexing of network layer protocols.**
 - **Error detection and flow control.**
- **Usage: Works with all IEEE 802 LAN standards (Ethernet, Token Ring, Wi-Fi).**

CSMA (Carrier Sense Multiple Access)

CSMA is a media access control method where each station listens to (senses) the channel before transmitting data.

The goal is to minimize collisions in a shared communication medium.

Working Principle

- 1. Carrier Sense – Before sending, a station checks if the channel is idle.**
- 2. Multiple Access – All stations have equal rights to access the medium.**
- 3. Transmission Decision:**
 - **If the channel is idle, transmit immediately.**
 - **If the channel is busy, wait for it to become idle.**

Types of CSMA:

1. 1-persistent CSMA

- **If the channel is idle → transmit immediately.**
- **If busy → continuously sense until idle, then transmit.**
- **High chance of collisions if multiple stations are waiting.**

2. Non-persistent CSMA

- **If busy → wait for a random backoff time before sensing again.**
- **Lower collisions, but longer delays.**

3. p-persistent CSMA (for slotted channels)

- **If idle → transmit with probability p , or wait for the next slot with probability $1-p$.**

Limitation of CSMA

- Collisions can still happen because of propagation delay (two stations might sense an idle channel and transmit almost simultaneously).

CSMA/CD (Carrier Sense Multiple Access with Collision Detection)

CSMA/CD is an improved version of CSMA where stations detect collisions during transmission and stop immediately to reduce wastage.

Working Steps

1. **Sense the channel** – Wait until idle.
2. **Transmit data.**
3. **Collision Detection** – While transmitting, the station monitors the channel.
 - If the signal on the channel differs from the transmitted signal → collision detected.
4. **Abort Transmission** – Stop sending immediately.
5. **Send Jam Signal** – Notify all stations about the collision.
6. **Backoff Algorithm** – Wait for a random time (using **Binary Exponential Backoff**) before retrying.

Advantages over CSMA

- Wastes less time: stops transmission immediately after collision.
- Fairer access due to backoff.

Where Used

- Used in **wired Ethernet (IEEE 802.3)** until switched Ethernet replaced shared mediums.
- Not suitable for **wireless networks** → collision detection is difficult due to signal fading, so Wi-Fi uses **CSMA/CA**.

CSMA/CA (Carrier Sense Multiple Access with Collision Avoidance)

Purpose: Prevent collisions before they happen, especially in wireless networks.

Why needed for wireless?

- In wireless, you can't reliably detect collisions (you can't listen to the channel while sending).
- Also, **hidden node problem** — two devices can't hear each other but can still collide at the receiver.

How it works:

1. **Carrier Sense** – Listen before transmitting.
2. **If idle** → Wait for a short period (Interframe Space, IFS) → then send.
3. **If busy** → Wait until idle, then start a **random backoff timer**.
4. **Collision Avoidance Mechanisms:**
 - **RTS/CTS handshake:**
 - Sender sends **Request to Send (RTS)**.
 - Receiver replies with **Clear to Send (CTS)**.
 - Only then the sender transmits data.
 - Helps prevent hidden node collisions.
5. **Acknowledgment (ACK)** after successful reception.

Feature	CSMA/CD	CSMA/CA
Full form	Carrier Sense Multiple Access with Collision Detection	Carrier Sense Multiple Access with Collision Avoidance
Collision Handling	Detects collisions after they happen	Tries to avoid collisions before they happen

Used in	Wired LAN (Ethernet)	Wireless LAN (Wi-Fi, IEEE 802.11)
Extra Mechanism	Jam signal, backoff	Backoff, RTS/CTS, ACK
Efficiency	Higher in wired	Better for wireless
Hidden Node Problem	Not present	Handled using RTS/CTS

ALOHA Protocol

ALOHA is a **random access protocol** for controlling how multiple devices share a single communication channel without centralized control.

1. Origin

- Developed in **1970** at the **University of Hawaii** by Norman Abramson.
- Initially used for wireless communication between islands.
- Inspired many later protocols (Slotted ALOHA, CSMA, Ethernet, Wi-Fi).

2. Purpose

- To allow multiple users to transmit over a shared medium **without prior coordination**.
- Minimize collisions and improve channel utilization.

3. Key Concepts

- **Vulnerable period:** Time during which a frame is susceptible to collision.
- **Throughput (S):** Successful transmission rate.
- **Offered load (G):** Average number of transmission attempts per frame time.

Types of ALOHA

Pure ALOHA

- **Idea:** Transmit whenever you have data.
- If a collision occurs (two or more stations transmit at the same time):
 - Sender waits for a **random time** and retransmits.
- **Vulnerable Period:** $2 \times$ frame time (time in which collision can occur).
- **Efficiency:** Max Efficiency=18.4% (≈ 0.184)
- **Advantage:** Simple.
- **Disadvantage:** Very low efficiency for high traffic.

Slotted ALOHA

- **Improvement** over Pure ALOHA.
- Time is divided into **equal slots**.
- Transmission can only start **at the beginning of a slot**.
- Collisions are reduced because stations align transmissions.
- **Vulnerable Period:** $1 \times$ frame time.
- **Efficiency:** Max Efficiency=36.8% (≈ 0.368)
- **Advantage:** Better efficiency than Pure ALOHA.
- **Disadvantage:** Requires time synchronization.

Feature	Pure ALOHA	Slotted ALOHA
Time Slots	No	Yes
Vulnerable Period	$2 \times \text{frame time}$	$1 \times \text{frame time}$
Max Efficiency	18.4%	36.8%
Complexity	Simple	Slightly complex (sync needed)
Performance	Lower	Higher

Limitations

- Works well only in **low traffic** conditions.
- Efficiency is low compared to modern MAC protocols.
- Collisions waste channel bandwidth.
- Slotted ALOHA requires **synchronization**.

Applications

- Early satellite communication systems.
- Wireless LAN research prototypes.
- Base for protocols like **Ethernet (CSMA/CD)** and **Wi-Fi (CSMA/CA)**.

Multiple Access Protocols and Networks

- **Multiple Access:** A set of techniques that allow multiple devices (stations/nodes) to share the same communication channel efficiently without interference.
- **Used in both wired and wireless communication.**
- **Goal: Maximize channel utilization, minimize collisions, and ensure fairness.**

Why Multiple Access is Needed

- In a network, a single communication medium (like cable, satellite link, or radio frequency band) is often shared.
- **Without a proper access method:**
 - **Multiple transmissions may collide.**
 - **Data may get lost.**
 - **Network performance drops.**

Classification of Multiple Access Protocols

Multiple Access Protocols can be classified into three main categories:

A. Random Access Protocols

- **No central control; stations transmit whenever they want.**
- **Collisions may occur → need collision handling.**
- **Used in bursty traffic conditions.**

Common protocols:

ALOHA

Pure ALOHA:

- **Station transmits immediately when data is ready.**
- **If collision → wait random time → retransmit.**
- **Max efficiency: 18.4%.**

Slotted ALOHA:

- **Time is divided into slots; send only at slot boundaries.**
- **Collisions reduced.**
- **Max efficiency: 36.8%.**

CSMA (Carrier Sense Multiple Access)

- **Station listens** before transmitting.
- If channel is busy → wait.
- **Variants:**
 - **1-persistent:** Transmit immediately when channel free.
 - **Non-persistent:** Wait random time before checking again.
 - **p-persistent:** Transmit with probability p .

CSMA/CD (Collision Detection)

- **Used in traditional Ethernet.**
- **Detects collisions during transmission.**
- **Stops sending immediately → reduces wasted time.**

CSMA/CA (Collision Avoidance)

- **Used in Wi-Fi.**
- **Avoids collisions using ACKs, RTS/CTS handshakes, and backoff timers.**

Pros of Random Access:

- **Simple implementation.**
- **Low delay when few users active.**

Cons:

- **Collisions waste bandwidth.**
- **Poor efficiency in high load.**

B. Controlled Access Protocols

- **Centralized or distributed control decides which station can transmit.**
- **No collisions (if implemented correctly).**
- **Used where predictable performance is needed.**

Methods:

1. Reservation:

- **Time slots are reserved for specific stations.**
- **Example: Reservation ALOHA.**

2. Polling:

- **Master station asks each station in turn.**
- **Example: Token Bus polling.**

3. Token Passing:

- **A special frame (token) circulates.**
- **Only the station with the token can transmit.**
- **Example: Token Ring, FDDI.**

Advantages:

- **Efficient under heavy load.**

Disadvantages:

- **More complex, possible single point of failure.**

C. Channelization Protocols

- **Channel is divided into smaller pieces in time, frequency, or code.**
- **Stations are assigned specific portions of the channel.**

Types:

1. FDMA (Frequency Division Multiple Access):

- Each station gets a different frequency band.
- Used in radio broadcasting, early cellular systems.

2. TDMA (Time Division Multiple Access):

- Each station gets a fixed time slot in a repeating cycle.
- Used in GSM networks.

3. CDMA (Code Division Multiple Access):

- All stations use the same frequency but with unique codes.
- Used in 3G mobile systems.

Advantages:

- No collisions.

Disadvantages:

- Less flexible for dynamic traffic.

Category	Examples	Collisions	Efficiency under load	Complexity
Random Access	ALOHA, CSMA/CD, CSMA/CA	Yes	Low	Low
Controlled Access	Polling, Token Passing	No	High	Medium
Channelization	FDMA, TDMA, CDMA	No	High	High

Protocol	Max Efficiency	Notes
Pure ALOHA	18.4%	High collisions at load.
Slotted ALOHA	36.8%	Better than Pure ALOHA.
CSMA	~50-90%	Depends on sensing method.
CSMA/CD	90%+	High efficiency on short links.
Controlled	~100%	When overhead is low.

Real-World Networks

- Ethernet (802.3): CSMA/CD (now mostly switched).
- Wi-Fi (802.11): CSMA/CA.
- GSM: TDMA.
- 3G: CDMA.
- 4G/5G: OFDMA.
- Satellite: ALOHA, TDMA.

Networks Layer Functions and Protocols

- The **Network Layer** is the **3rd layer** in the OSI model (above Data Link Layer, below Transport Layer).
- Main responsibility: **host-to-host delivery** of packets **across multiple networks**.
- Works with **logical addresses** (e.g., IP addresses) instead of physical (MAC) addresses.
- This layer ensures that data can **find its way** from the source to the destination, even through multiple intermediate devices (routers).

Key Functions of the Network Layer

1. Logical Addressing

- Provides a **unique address** for each device on the network.
- Physical addresses (MAC) are only valid within the same local network.
- Logical addresses (e.g., IPv4: 32 bits, IPv6: 128 bits) are hierarchical and help identify network + host.

2. Routing

- Determines the **optimal path** for packets to travel from source to destination.
- Uses routing algorithms (shortest path, distance vector, link state).
- Performed by routers using routing tables.
- Can be static (manually configured) or dynamic (updated by routing protocols).

3. Packet Forwarding

- **The actual process of moving packets from an incoming link to the correct outgoing link.**
- **Based on destination IP address and routing table entries.**

4. Fragmentation and Reassembly

- **If a packet is too large for the next network's Maximum Transmission Unit (MTU):**
 - **Fragmentation: Break packet into smaller fragments.**
 - **Reassembly: Destination host combines fragments back into the original packet.**
- **IPv4 supports fragmentation; IPv6 avoids fragmentation by source devices determining MTU.**

5. Error Handling and Diagnostics

- **Reports errors (e.g., unreachable host, TTL expired) using ICMP (Internet Control Message Protocol).**
- **Example: ping uses ICMP Echo Request/Reply.**

6. Congestion Control

- **Prevents network overload by controlling the amount of data entering the network.**
- **Uses techniques like packet dropping, choke packets, or routing changes.**

7. Quality of Service (QoS)

- Prioritizes certain types of traffic (e.g., VoIP, streaming) for better performance.
- Implemented via mechanisms like Differentiated Services (DiffServ).

Network Layer Protocols

Protocol	Description	Key Features
IPv4 (Internet Protocol version 4)	Core protocol of TCP/IP	32-bit addresses, supports fragmentation, no encryption.
IPv6	Next-gen IP	128-bit addresses, no fragmentation, better security (IPsec), simplified header.
ICMP	Error reporting & diagnostics	Sends control messages (e.g., Destination Unreachable, Echo).
IGMP	Internet Group Management Protocol	Manages multicast group memberships.
ARP (Address Resolution Protocol)	Maps IPv4 address to MAC	Works between Network & Data Link layers.
RARP	Reverse ARP	Maps MAC to IPv4 address (obsolete).

Routing Protocols

- **Distance Vector: RIP** (Routing Information Protocol) — periodic updates with hop counts.
- **Link State: OSPF** (Open Shortest Path First) — builds full network map.
- **Path Vector: BGP** (Border Gateway Protocol) — used between ISPs.

1. Distance Vector (DV) Routing Protocols

- **Each router advertises its routing table to neighbors periodically.**
- **Information contains distance metric (e.g., hop count) to each destination.**
- **Uses Bellman–Ford algorithm.**
- **Advantages:**
 - **Simple to implement.**
 - **Low computational load.**
- **Disadvantages:**
 - **Slow convergence.**
 - **Prone to routing loops.**
- **Examples:**
 - **RIP (Routing Information Protocol) — Max hop count: 15.**
 - **IGRP (Interior Gateway Routing Protocol — Cisco proprietary).**

2. Link State (LS) Routing Protocols

- **Each router discovers full network topology by exchanging link-state advertisements (LSAs).**
- **Builds a complete map of the network and calculates the best path using Dijkstra's shortest path algorithm.**
- **Advantages:**
 - **Fast convergence.**
 - **Less prone to loops.**
- **Disadvantages:**
 - **Higher memory and CPU usage.**
 - **More complex.**

- **Examples:**
 - **OSPF (Open Shortest Path First)** — open standard.
 - **IS-IS (Intermediate System to Intermediate System).**

3. Path Vector Routing Protocols

- **Used for inter-domain routing (between different autonomous systems).**
- **Each route update includes the full path (list of AS numbers).**
- **Prevents loops by rejecting routes containing its own AS number.**
- **Example:**
 - **BGP (Border Gateway Protocol)** — backbone of the Internet.

RIP (Routing Information Protocol)

- **Distance Vector, uses hop count.**
- **Max hops = 15; beyond that → unreachable.**
- **Updates every 30 seconds.**
- **Uses split horizon and poison reverse to avoid loops.**

OSPF (Open Shortest Path First)

- **Link State protocol.**
- **Uses areas to scale large networks.**
- **Supports VLSM and CIDR.**
- **Fast convergence.**

BGP (Border Gateway Protocol)

- **Path Vector protocol, uses TCP (port 179) for reliable updates.**
- **Internet-scale routing.**
- **Policy-based routing (not always shortest path — can be based on business agreements).**

Devices Working at Network Layer

- **Router — main device, forwards packets based on IP.**
- **Layer 3 Switch — can perform routing + switching.**
- **Firewall (Layer 3) — can filter traffic based on IP addresses.**

1. Router

- **Primary Layer 3 device that connects different networks.**
- **Reads IP addresses from incoming packets and determines the best path.**
- **Functions:**
 - **Routing (static & dynamic).**
 - **Packet forwarding.**
 - **NAT (Network Address Translation).**
 - **ACL (Access Control Lists) for security filtering.**
- **Examples: Cisco ISR Series, Juniper MX Series.**

2. Layer 3 Switch

- **Combines features of a switch (Layer 2) and a router (Layer 3).**
- **High-speed packet forwarding within VLANs and routing between VLANs.**

- **Functions:**
 - **Inter-VLAN routing.**
 - **Supports routing protocols (OSPF, EIGRP).**
 - **Often used in enterprise LAN backbones.**
- **Example: Cisco Catalyst 3560, HP ProCurve.**

3. Multilayer Switch

- **Works at Layer 2, Layer 3, and sometimes Layer 4.**
- **Can make routing decisions based on IP and transport layer info (e.g., TCP/UDP ports).**
- **Use case: Large data centers for both switching and policy-based routing.**

4. Firewall (Layer 3-capable)

- **Security device that can operate at multiple layers.**
- **At Layer 3, it filters traffic based on IP addresses, subnets, or routing rules.**
- **Can perform packet inspection, NAT, and VPN tunneling.**
- **Examples: Palo Alto Networks, Cisco ASA, Fortinet.**

5. Layer 3 Load Balancer

- **Distributes network traffic across multiple servers using IP-based routing decisions.**
- **Works at Layer 3 to direct packets based on destination IP or routing policies.**
- **Example: F5 BIG-IP (can also work at Layer 4/7).**

6. Gateway (Layer 3 function)

- **Connects networks using different protocols or architectures.**
- **Performs protocol translation (e.g., IPv4 ↔ IPv6, or IP ↔ Non-IP networks).**
- **Can be software-based or hardware-based.**

Supporting Components

- **Routing Table: Maintained in all Layer 3 devices, maps destinations to next hops.**
- **Forwarding Information Base (FIB): Optimized table for fast packet forwarding.**
- **Control Plane & Data Plane:**
 - **Control plane runs routing protocols and updates tables.**
 - **Data plane moves packets according to the tables.**

Transport Layer Functions and Protocols

The Transport Layer is the 4th layer in the OSI model (above the Network Layer and below the Session Layer).

Its main responsibility is to provide reliable, transparent transfer of data between end systems (hosts) and to ensure that the data arrives error-free, in sequence, and with no losses or duplications.

In the **TCP/IP model**, the Transport Layer sits above the Internet Layer and supports **end-to-end communication** between applications running on different hosts.

Key Characteristics

- **End-to-End Communication:** Establishes communication directly between processes (applications), not just between hosts.
- **Segmentation and Reassembly:** Breaks messages into smaller segments and reassembles them at the destination.
- **Error Detection and Recovery:** Ensures reliability through acknowledgments and retransmissions.
- **Flow Control:** Prevents the sender from overwhelming the receiver.
- **Multiplexing:** Supports multiple communication sessions over a single network link.

Functions of the Transport Layer

1. Process-to-Process Delivery

- While the Network Layer delivers packets from one host to another (host-to-host), the Transport Layer delivers data from one specific process to another.
- Achieved via port numbers.
 - Example: HTTP uses port 80, HTTPS uses port 443.

2. Addressing (Port Numbers)

- Port numbers identify sending and receiving processes.
- Two types:
 - **Well-known ports (0–1023):** Assigned to common protocols (e.g., 21 for FTP, 25 for SMTP).
 - **Ephemeral ports:** Dynamically assigned for client-side communication.

3. Segmentation and Reassembly

- **Data from the application layer is split into smaller chunks called segments.**
- **Each segment contains a header with sequence numbers for reassembly.**
- **The receiver uses these numbers to reorder segments if they arrive out of sequence.**

4. Connection Control

- **Connection-oriented: Before sending data, a connection is established (TCP).**
- **Connectionless: Data is sent without prior connection setup (UDP).**

5. Flow Control

- **Prevents a fast sender from overwhelming a slow receiver.**
- **Common methods:**
 - **Sliding Window Protocol**
 - **Credit-based flow control**
 - **Example: TCP uses window size in its header.**

6. Error Control

- **Ensures the delivery of error-free data.**
- **Uses:**
 - **Checksums (error detection).**
 - **Acknowledgments (ACK).**
 - **Retransmissions when errors are detected.**

7. Multiplexing and Demultiplexing

- **Multiplexing: Multiple processes share the same network connection.**
- **Demultiplexing: The receiver identifies which process the data belongs to by reading the port number in the segment.**

8. Quality of Service (QoS)

- **Some transport protocols can prioritize data (important in multimedia streaming, VoIP).**
- **Example: Differentiated Services in TCP/IP.**

9. End-to-End Reliability

- Transport layer maintains reliability between end hosts, independent of the reliability of the underlying network.

Transport Layer Protocols

1. Transmission Control Protocol (TCP)

- Connection-oriented, reliable transport protocol.
- Main Features:
 - Three-way handshake for connection establishment.
 - Segmentation and sequencing.
 - Error detection and retransmission.
 - Flow control using sliding windows.
 - Full-duplex communication.

TCP Header Format (20–60 bytes):

Field	Size	Description
Source Port	16 bits	Sending process
Destination Port	16 bits	Receiving process
Sequence Number	32 bits	Byte number of first byte in segment
Acknowledgment Number	32 bits	Next byte expected
Data Offset	4 bits	Header length
Flags	6 bits	Control bits (SYN, ACK, FIN, RST, URG, PSH)
Window Size	16 bits	Flow control
Checksum	16 bits	Error detection
Urgent Pointer	16 bits	URG flag handling
Options	Variable	Additional features

2. User Datagram Protocol (UDP)

- Connectionless, unreliable, lightweight transport protocol.
- Main Features:

- **No handshake.**
- **No sequencing or acknowledgment.**
- **Minimal overhead (8-byte header).**
- **Faster than TCP, suitable for real-time applications.**

UDP Header Format (8 bytes):

Field	Size	Description
Source Port	16 bits	Sending process
Destination Port	16 bits	Receiving process
Length	16 bits	Length of UDP header + data
Checksum	16 bits	Error detection

3. Stream Control Transmission Protocol (SCTP)

- **Combines features of TCP and UDP.**
- **Main Features:**
 - **Connection-oriented.**
 - **Multi-streaming (multiple streams in one connection).**
 - **Multi-homing (multiple network paths).**
 - **Message-oriented (like UDP).**
- **Commonly used in telecom signaling (SS7 over IP).**

TCP vs UDP vs SCTP:

Feature	TCP	UDP	SCTP
Connection	Oriented	Connectionless	Oriented
Reliability	Yes	No	Yes
Sequencing	Yes	No	Yes
Flow Control	Yes	No	Yes
Overhead	High	Low	Medium

Speed	Slower	Faster	Medium
Use Case	Web, Email, FTP	VoIP, Streaming	Telephony, Signaling

Common Transport Layer Services

1. **Reliable data transfer.**
2. **Connection management.**
3. **Error detection and recovery.**
4. **Data segmentation.**
5. **Port addressing.**
6. **Flow control.**
7. **Congestion control (TCP-specific).**

1. Reliable Data Transfer

- **Definition:**
The Transport Layer ensures that the data sent by the sender is delivered to the receiver without errors, loss, duplication, or out-of-order delivery.
- **How it's achieved:**
 1. **Error Detection** → Using checksums in the segment header.
 2. **Acknowledgments (ACKs)** → Receiver confirms correct segment reception.
 3. **Retransmission** → Lost or corrupted segments are resent.
 4. **Sequencing** → Sequence numbers keep data in correct order.
- **Example:**
TCP ensures reliability by:
 - Sending segments in sequence.
 - Resending lost packets.
 - Reordering out-of-order packets.

2. Connection Management

- **Definition:**

The process of establishing, maintaining, and terminating a connection between two devices before and after data transfer.

- **Types:**

- **Connection-Oriented (e.g., TCP):**

- Requires a setup phase (3-way handshake) before sending data.
 - Example: Web browsing (HTTP over TCP).

- **Connectionless (e.g., UDP):**

- No setup phase; packets sent directly.
 - Example: Streaming, DNS queries.

- **TCP Example:**

SYN → Client to Server (request to connect).

SYN-ACK → Server to Client (acknowledgment + request).

ACK → Client to Server (final confirmation).

3. Error Detection and Recovery

- **Definition:**

Identifying and correcting errors in transmitted data.

- **Techniques:**

- **Checksums** → Verify integrity of the segment.

- **ACK/NACK (Negative ACK)** → Indicates which data needs retransmission.

- **Timeouts** → If no ACK received within a time frame, retransmit.

- **Protocols:**

- **TCP** → Provides error recovery.

- **UDP** → Detects errors but doesn't correct them (application handles it).

4. Data Segmentation and Reassembly

- **Segmentation:**
 - Application data may be too large for network transmission.
 - Transport Layer splits it into segments (TCP) or datagrams (UDP).
 - Each segment has a header containing sequence and control information.
- **Reassembly:**
 - Receiver uses sequence numbers to reconstruct original message.
 - If segments arrive out of order, they are reordered before delivering to the application.
- **Example:**
A 5 MB file sent via TCP → broken into multiple 1 KB segments → reassembled by the receiver.

5. Port Addressing

- **Definition:**
Identifying the sending and receiving processes using port numbers.
- **Purpose:**
 - Multiple applications on the same device can send/receive data simultaneously.
- **Port Types:**
 - Well-known ports (0–1023): Fixed for standard services (e.g., HTTP → 80, FTP → 21).
 - Registered ports (1024–49151): Assigned to specific apps.
 - Ephemeral/Dynamic ports (49152–65535): Temporary client-side ports.
- **Example:**
 - You can browse the web (port 80), download files via FTP (port 21), and check email (port 110) simultaneously.

6. Flow Control

- **Definition:**
A mechanism to ensure that the sender does not overwhelm the receiver by sending too much data too quickly.

- **How TCP handles it:**
 - **Sliding Window Protocol:**
 - The sender can only send as many bytes as allowed by the receiver's advertised window size.
 - The window slides forward as ACKs are received.
- **Example:**

If receiver's buffer can store 5 KB at a time → TCP adjusts sending rate accordingly.

7. Congestion Control

- **Definition:**

Preventing excessive traffic in the network that could cause packet loss and delays.
- **TCP's Congestion Control Techniques:**
 1. **Slow Start** → Start with small window, increase gradually.
 2. **Congestion Avoidance** → Increase window linearly after a threshold.
 3. **Fast Retransmit** → Retransmit lost packet immediately when multiple duplicate ACKs are received.
 4. **Fast Recovery** → Avoid going back to slow start unnecessarily.
- **Example:**

Streaming video may slow down temporarily if congestion is detected.

Examples in the Real World

- **Web Browsing (HTTP/HTTPS)** → TCP (port 80/443)
- **Video Streaming (YouTube, Netflix)** → Mostly UDP (low-latency streaming)
- **Voice Calls (VoIP)** → UDP (real-time)
- **Email (SMTP, POP3, IMAP)** → TCP
- **Telecom signaling** → SCTP

Application Layer

- **Application Layer is the topmost layer in the OSI model and part of the Application Layer in TCP/IP.**
- **It provides services directly to end users or applications to enable communication over the network.**
- **Purpose:**
 - **Facilitate user-level network communication.**
 - **Ensure the right protocols and formats are used for data exchange.**
 - **Hide lower-layer complexities from the user.**

2. Role of Application Layer Protocols

- **Define rules and formats for exchanging data between applications.**
- **Provide interfaces for software to access network services.**
- **Support multiple communication models:**
 - **Client-Server (e.g., HTTP, FTP)**
 - **Peer-to-Peer (e.g., BitTorrent)**

3. Key Functions

1. Identifying Communication Partners

- **Determines who to communicate with (e.g., IP + Port mapping).**

2. Determining Resource Availability

- **Checks if the requested service or file is available.**

3. Synchronizing Communication

- **Ensures both sides follow the same message format and sequence.**

4. Data Formatting

- **Converts application data into a standard format (e.g., HTML, XML).**

5. Error Handling

- **Handles application-level errors like missing files or incorrect commands.**

Application Layer Protocols

Protocol	Full Form	Purpose	Port
HTTP/HTTPS	HyperText Transfer Protocol (Secure)	Web page transfer	80 / 443
FTP	File Transfer Protocol	Transfer files between computers	21
SMTP	Simple Mail Transfer Protocol	Send emails	25
POP3	Post Office Protocol v3	Receive emails (download)	110
IMAP	Internet Message Access Protocol	Receive emails (sync)	143
DNS	Domain Name System	Convert domain names to IP addresses	53
TELNET	Telecommunication Network	Remote login	23
SSH	Secure Shell	Secure remote login	22
SNMP	Simple Network Management Protocol	Manage network devices	161
DHCP	Dynamic Host Configuration Protocol	Assign IP addresses dynamically	67/68

Categories of Application Layer Protocols

1. Web Communication Protocols

- HTTP, HTTPS, WebSocket.

2. File Transfer Protocols

- FTP, SFTP, TFTP.

3. Email Protocols

- SMTP, POP3, IMAP.

4. Network Management Protocols

- SNMP, Syslog.

5. Remote Access Protocols

- **Telnet, SSH.**

6. Directory Services

- **LDAP, NIS.**

7. Name Resolution Protocols

- **DNS.**

8. Dynamic Addressing Protocols

- **DHCP.**

HTTP & HTTPS

- **Purpose, features, stateless nature.**
- **GET/POST methods.**
- **HTTPS working with TLS/SSL.**
- **Example: How a browser loads a web page.**
- **Pros & cons.**

FTP

- **Active vs passive mode.**
- **Commands and replies.**
- **Authentication process.**
- **Use cases.**

SMTP, POP3, IMAP

- **How SMTP sends an email.**
- **Difference between POP3 and IMAP.**
- **Diagram of full email cycle.**

DNS

- **Purpose, hierarchy (Root, TLD, authoritative).**
- **Query types (Recursive vs Iterative).**
- **DNS cache and resolver working.**

DHCP

- **Four-step process (DORA: Discover, Offer, Request, Acknowledge).**
- **Lease renewal.**
- **Advantages & disadvantages.**

SNMP

- **Components (Manager, Agent, MIB).**
- **Versions (v1, v2c, v3).**
- **Security considerations.**

SSH

- **Secure authentication.**
- **Encryption process.**
- **Differences from Telnet.**

Advantages of Application Layer

1. Direct User Interaction

- **Provides services that the end user can directly use (e.g., web browsing, email).**

2. Protocol Variety

- **Supports multiple protocols for different purposes (HTTP, FTP, DNS, etc.).**

3. Hardware & OS Independent

- **Works the same across platforms.**

4. Data Formatting & Translation

- **Converts data into formats understandable by applications.**

5. Integration with Lower Layers

- **Works with TCP/UDP for reliable or fast communication.**

6. Extensibility

- **New protocols can be added easily without changing hardware.**

Disadvantages of Application Layer

1. Security Risks

- **Vulnerable to application-level attacks like phishing, malware, and XSS.**

2. Performance Overhead

- **Some protocols (e.g., HTTPS) require encryption, increasing processing time.**

3. Dependency on Lower Layers

- **If transport or network layers fail, application services stop.**

4. Protocol Complexity

- **Different protocols require different configurations and understanding.**

5. Possible Compatibility Issues

- **Older systems may not support modern protocol versions.**

Advantages & Disadvantages of Key Application Layer Protocols

1. HTTP / HTTPS

Advantages:

- **Widely supported and easy to implement.**
- **HTTPS provides encryption for security.**
- **Stateless nature allows scalability.**

Disadvantages:

- **Statelessness can require extra overhead (cookies, sessions).**
- **HTTP without encryption is insecure.**

2. FTP

Advantages:

- **Simple method for file transfer.**
- **Supports both binary and text file transfers.**

Disadvantages:

- **Plain FTP is not secure (credentials sent in plain text).**
- **Less efficient for small, frequent transfers.**

3. SMTP

Advantages:

- **Standardized for sending emails.**
- **Works well with other email protocols.**

Disadvantages:

- **Only for sending, not receiving.**
- **Can be exploited for spam without security measures.**

4. POP3

Advantages:

- **Simple protocol for downloading emails.**
- **Emails can be read offline.**

Disadvantages:

- **Emails are usually deleted from server after download.**
- **Poor for multi-device access.**

5. IMAP

Advantages:

- **Synchronizes emails across devices.**
- **Allows partial downloading for faster access.**

Disadvantages:

- **Requires constant server connection.**
- **More complex than POP3.**

6. DNS

Advantages:

- **Makes internet navigation easy using domain names.**
- **Distributed database ensures redundancy.**

Disadvantages:

- **Vulnerable to DNS spoofing and cache poisoning.**
- **Queries are often unencrypted.**

7. DHCP

Advantages:

- **Automatic IP assignment reduces admin work.**
- **Prevents manual configuration errors.**

Disadvantages:

- **Not secure — IPs can be hijacked.**
- **If DHCP server fails, clients may not get IPs.**